

Informed Consent Form

Online Survey

Hawaii Pacific University

Principal Investigator: Dr. Mary L. Smith

Department: Computer Science and Engineering

Study Title: NSF CDER CyberTraining Grant: Modern Course Exemplars infused with Parallel and Distributed Computing for CS1 & CS2

IRBNet ID: ###

You are invited to participate in an online research study on **improving students' knowledge of parallel programming and distributed computing in CSCI 2911-Computer Science I and CSCI 2912 - Computer Science II**. This study is being conducted by **Dr. Mary L. Smith**, Associate Professor of Computer Science at the College of Natural and Computational Sciences

Please read this form and ask any questions you may have before agreeing to be a part of the study.

What will happen if I participate?

Your participation will take approximately 10 minutes of your time. You will be asked to complete an online survey about your prior experience on programming fundamentals. It is ok to answer, "No experience at all". The instructor would give this answer for some of these questions, and you can succeed in this course without having seen any of these topics. We just ask broadly to capture all possible kinds of prior experience. By the end of the semester, in the last week of classes, you will be asked to take the survey again as a post survey of your experience on programming fundamentals.

What are my options?

Your role as a participant in this study will include allowing de-identified data from several surveys/reports to be used in study results published to the computer science community. All information taken from the study will be coded to protect your name. No personally

identifiable information will be included with your data and your identity will be fully protected. Once the data has been fully analyzed, it will be destroyed. Your decision to participate is completely voluntary, and you have the right to terminate your participation without penalty. If you do not wish to complete this survey, simply select "I do not consent" and close the survey. The Principal Investigator of this study, Dr. Mary L. Smith, can be contacted via email at mlsmith@hpu.edu, if you should have any questions about this research study.

How will my information be used?

Efforts will be made to keep your personal information confidential. Absolute confidentiality cannot be guaranteed. All information taken from the study will be coded to protect your name. No personally identifiable information will be included with your data and your identity will be fully protected. Once the data has been fully analyzed, it will be destroyed. Your identity will not be shared if the study findings are published or presented. Only Dr. Smith will have access to the collected data. Any data that will be disseminated to other institutions or the National Science Foundation will be completely coded, anonymized, and deidentified. All collected information will be stored in a cloud drive maintained and protected under HPU's policies.

What are my risks?

There are no risks to individuals participating in this survey beyond those that exist in daily life.

May I be eligible for compensation?

No compensation is being offered to the participant.

Who can answer my questions about this study?

If you have questions about the study or encounter a problem with the research, contact the researcher, Dr. Mary L. Smith at mlsmith@hpu.edu.

The research team members are not acting as agents of the Hawaii Pacific University and do not have authority to bind the University. Any opinions, findings, or recommendations expressed in this study represent the researchers' own and do not necessarily reflect the views of the Hawaii Pacific University.

- ☐ I am 18 years of age or older and I consent to participate in the study
- ☐ I do not consent, I do not wish to participate, or I am under 18 years of age

Block 2

Most of the questions below ask about your previous experience. It is ok to answer "No experience at all". The instructor would give this answer for some of these questions and you can succeed in this course without having seen any of these topics. We just ask broadly in order to capture all possible kinds of prior experience.

CSCI 2911 Fall 2024 Pre-Course Experience Survey

Time Requirement: under 10 minutes

Purpose: We use this pre-course experience survey for the purpose of customizing content, aiding in engagement and motivation, managing class time efficiently, implementing differentiation strategies, and building on prior knowledge

Rate your computer/programming experience:

It is ok to answer, "Not Experienced At All." You can succeed in this course without prior experience.

	Not experienced at all	Slightly experienced	Moderately experienced	Very experienced	Extre experie
Computer experience (internet searches, email, discord, discussion groups, games, word processing, spreadsheets, organizing files & folders, etc.)	☐	☐	☐	☐	☐
Programming Experience (self-initiated learning and/or formal programming course)	☐	☐	☐	☐	☐

	Not experienced at all	Slightly experienced	Moderately experienced	Very experienced	Extremely experienced
Programming Experience compared to your classmates	☐	☐	☐	☐	☐

Please rate your experience with the following programming languages:

	Not comfortable at all	Slightly comfortable	Moderately comfortable	Very comfortable	Extremely comfortable
C	☐	☐	☐	☐	☐
C++	☐	☐	☐	☐	☐
C#	☐	☐	☐	☐	☐
Java	☐	☐	☐	☐	☐
Javascript	☐	☐	☐	☐	☐
PHP	☐	☐	☐	☐	☐
Python	☐	☐	☐	☐	☐
Scratch	☐	☐	☐	☐	☐
Visual Basic	☐	☐	☐	☐	☐
Other	☐	☐	☐	☐	☐

Please rate your experience with using the following in your programs:

	No experience at all	Slightly experienced	Moderately experienced	Very experienced	Extremely experienced
Data Types (int, float, double, boolean, char)	☐	☐	☐	☐	☐
Arithmetic Expressions & Operators	☐	☐	☐	☐	☐
Branching (if/else, relational & logical operators, relational expressions)	☐	☐	☐	☐	☐
Loops (while, for)	☐	☐	☐	☐	☐
Calling functions/methods	☐	☐	☐	☐	☐
Arrays	☐	☐	☐	☐	☐

	No experience at all	Slightly experienced	Moderately experienced	Very experienced	Extremel experienc
Writing static functions	☐	☐	☐	☐	☐
Writing methods	☐	☐	☐	☐	☐
Writing classes					
Creating objects	☐	☐	☐	☐	☐

Where did you get your programming experience?

(select all that applies)

- ☐ CSCI 2911 (previous semester)
- ☐ AP Computer Science A
- ☐ AP Computer Science Principles
- ☐ Highschool Computer Science class (other than AP)
- ☐ Online course with an assigned instructor/teacher
- ☐ Duel credit course with another institution
- ☐ Online tutorial or websites (self-initiated)
- ☐ Other

Rate your experience with the following:

	Not experienced at all	Slightly experienced	Moderately experienced	Very experienced	Extremely experienced
Parallel processing: Computing where the operations within a program are being executed simultaneously (i.e., running on different cores of the processor at the same time)	☐	☐	☐	☐	☐
Distributed computing: Computing with a program that runs on multiple computers or	☐	☐	☐	☐	☐

Not
experienced
at all

Slightly
experienced

Moderately
experienced

Very
experienced

Extremely
experienced

that runs using data
from a different
computer (i.e., an
online service)

Event handling:

Having a function that
responds to an
independent action
(i.e., a mouse click)



What grade do you expect to get in CSCI 2911?

- ☐ A (90-100)
- ☐ B (80-89.9)
- ☐ C (70-79.9)
- ☐ D (60-69.9)
- ☐ F (0-59.9)

What specific skills or abilities do you aspire to acquire or improve upon as a result of your learning in this class?

HPU Email Address:

(only used to connect your responses at the beginning and end of the course)

Please indicate your ethnicity (optional)

- ☐ Asian / Pacific Islander
- ☐ American Indian or Alaskan Native
- ☐ Black / African American
- ☐ Hispanic
- ☐ White / Caucasian
- ☐ Two or more ethnicities

Please indicate your gender (optional)

- ☐ Male
- ☐ Female
- ☐ Other
- ☐ Prefer not to respond

Student Age

- ☐ 18-22
- ☐ 22-27
- ☐ 28-32
- ☐ 33 or over

When did you first begin attending HPU? (includes Fall, Spring, or Summer start time)

- ☐ Before 2020
- ☐ 2020-2021 academic year
- ☐ 2021-2022 academic year
- ☐ 2022-2023 academic year
- ☐ 2023-2024 academic year
- ☐ 2024-2025 academic year

What is your current class standing?

- ☐ First year (freshman)
- ☐ Sophomore
- ☐ Junior
- ☐ Senior
- ☐ Graduate student

What is your major? (select all that apply)

- ☐ Computer Science
- ☐ Biotechnology Engineering
- ☐ Biomedical Engineering
- ☐ Electrical Engineering
- ☐ Math
- ☐ Other